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Anti-Vaccine Activists Use A Federal Database To Spread Fear About COVID Vaccines

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HEARD ON MORNING EDITION

Geoff Brumfiel

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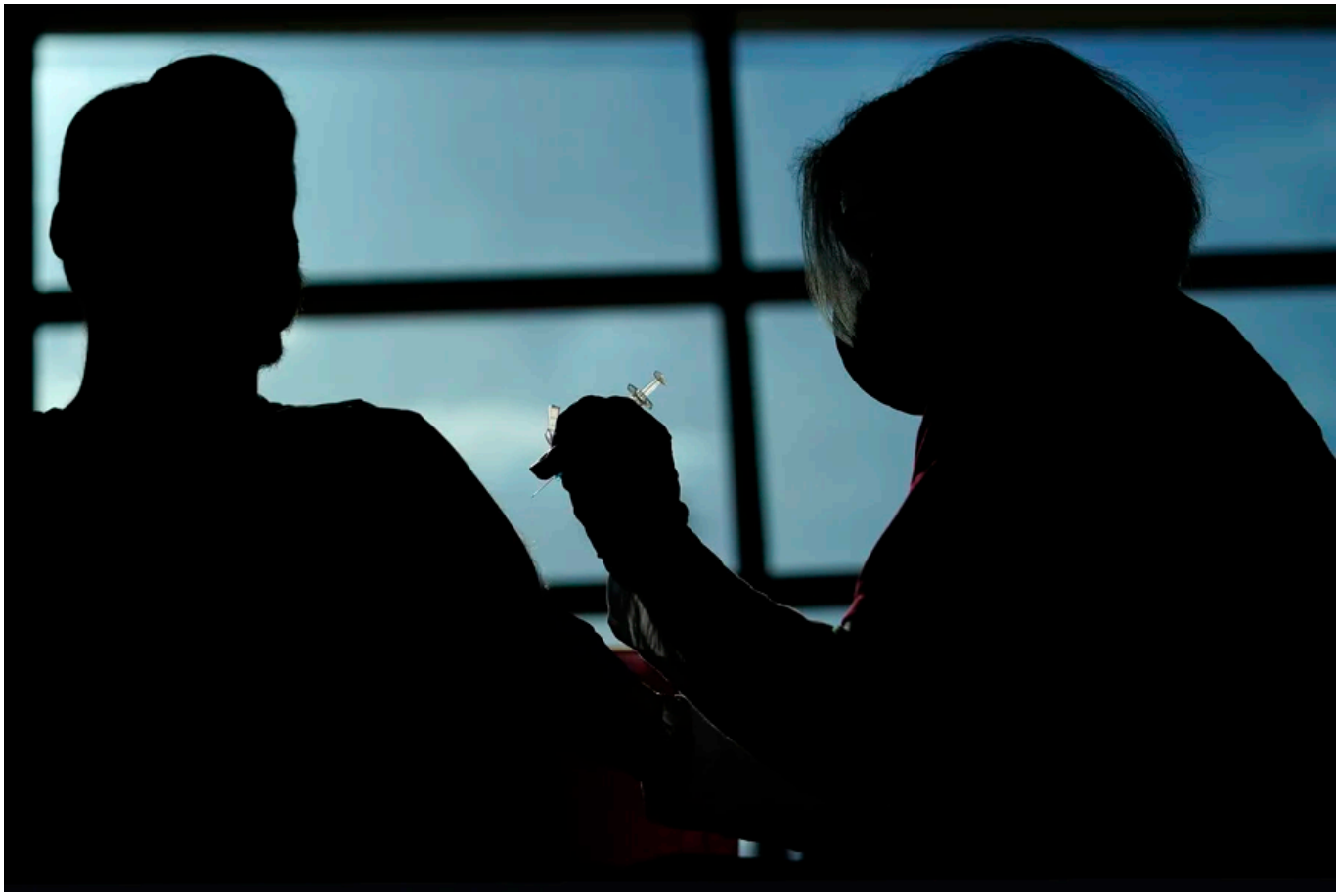

Solid research has found the vaccines authorized for use against COVID-19 to be safe and effective. But some anti-vaccine activists are mischaracterizing government data to imply the jabs are dangerous.

Matt Slocum/AP

The largest U.S. database for detecting events that might be vaccine side effects is being used by activists to spread disinformation about COVID-19 vaccines.

Known as the Vaccine Adverse Event Reporting System, or VAERS, the database includes hundreds of thousands of reports of health events that occurred minutes, hours or days after vaccination. Many of the reported events are coincidental — things that happen by chance, not caused by the shot. But when millions of people are vaccinated within a short period, the total number of these reported events can look big.

Epidemiologists consider this database as only a starting point in the search for rare but potentially serious vaccine side effects. Far more work must be done before a cause-and-effect link can be determined between a reported health event and a vaccine.

"It's a very valuable system for detecting adverse events, but it has to be used properly," says William Moss, executive director of the International Vaccine Access Center at the Johns Hopkins Bloomberg School of Public Health. "And it's ripe for misuse."

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In fact, VAERS has played a major role in the spread of misinformation about COVID-19 vaccines. The data is regularly appropriated by anti-vaccine advocates, who use the reports to claim falsely that COVID-19 vaccines are dangerous. They are aided by the fact that the entire VAERS database is public — it can be downloaded by anyone for any purpose.

"There's very little control over what can be accessed and what can be manipulated," says Melanie Smith, director of analysis at Graphika, a company that tracks vaccine misinformation online. She says that she sees VAERS data being shared across a wide variety of anti-vaccine social media channels. "I would say almost every mis- and disinformation story that we cover is accompanied by some set of VAERS data."

VAERS was established decades ago, partly in direct response to the anti-vaccine movement. In 1982, a TV documentary called *DPT Vaccine Roulette* aired nationwide. It was filled with unsubstantiated claims that the vaccine given at the time against diphtheria, pertussis and tetanus could lead to intellectual and physical disability.

SHOTS - HEALTH NEWS

For Some Anti-Vaccine Advocates, Misinformation Is Part Of A Business

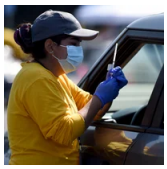
"It led to a massive number of lawsuits," says Dr. Walter Orenstein, associate director of the Emory Vaccine Center and a former director of the U.S. immunization program at the Centers for Disease Control and Prevention.

The legal battles got so bad that many pharmaceutical companies decided it wasn't worth making the vaccines. The U.S. began experiencing shortages. Congress stepped in with a law protecting manufacturers, and part of that act created VAERS in 1990. Right away, the database looked different from other collections of government medical data: Anybody could report a side effect from a vaccine (not just doctors), and anyone could request the entire VAERS database for any reason. Orenstein says the goal was to make it as open as possible.

"There were conspiracy theories, there were concerns that people were hiding things, and we didn't want to hide anything," he recalls. "It was very important that this system be publicly available so that others could look at it, and make their own conclusions if they didn't trust what the data were that the CDC and [Food and Drug Administration] were putting out."

Ever since, anti-vaccine groups have been using VAERS to push their unfounded theories about the dangers of vaccination. "VAERS data is often shared in the anti-vax community with the understanding that it's something that they fought for," Smith says. Since the rollout of the COVID-19 vaccines, Smith says that anti-vaccine advocates have been sharing YouTube videos showing how to download the data. Lately, she says infographics based on the data "seem to be really popular at the moment." They proliferate on alternative social media platforms such as Telegram.

The most commonly cited statistic among anti-vaccine groups is death following vaccination. Graphics from anti-vaccine proponents frequently tick off the number of deaths directly reported in VAERS — without noting the reports there have not been investigated or verified as causally linked to an immunization. Those numbers even made it onto the show of Fox News commentator Tucker Carlson last month. In a segment on the supposed dangers of COVID-19 vaccines, Carlson incorrectly claimed the system had recorded thousands of unexplained deaths. "It's clear that what is happening now is not even close to normal," he told his audience.



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Few Facts, Millions Of Clicks: Fearmongering Vaccine Stories Go Viral Online

HEALTH CARE

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The problem, says Saad Omer, director of the Yale Institute for Global Health, is that many of those deaths in the VAERS database were caused by other illnesses that happened around the same time as the immunization and had nothing to do with a vaccine: "Vaccines decrease your risk of COVID-19," Omer notes. "They don't make you immortal."

In fact, COVID-19 vaccines were given first to some of the oldest and sickest people in America. Their risk of dying from COVID was high, but "their risk of mortality due to other causes was also high. In fact, very high," Omer says.

He says it's not surprising that, after administering many millions of doses, a few thousand might coincidentally die soon after getting the vaccine. VAERS is where that data is recorded, and anti-vaccine campaigners then cite the number as people killed by the vaccine.

Meanwhile, Omer and his colleagues have done their own analysis and found the vaccines are saving quite a few people's lives. "We showed that there is a 99% reduction in mortality after two doses, and a 64% reduction in mortality even after one dose," he says.

Individual case reports in VAERS are also often cited as though they were studies of what can go wrong with vaccination, says Moss of Johns Hopkins. "This is really hard, because individual stories are really powerful," he says. But because of the system's openness, these anecdotes are unverified. In the early 2000s, an anesthesiologist falsely reported he had been turned into the Incredible Hulk by the flu shot, and the report appeared in VAERS. (It was later removed.)

"There's absolutely no screening," Moss says. Even if most reports are honest, they still don't come close to proving a causal link between a vaccine and a health event.

In an emailed statement, the CDC tells NPR the agency is aware of the misuse of the VAERS data but has no immediate plans to change the system. That's in part because VAERS is one of the agency's best sources for early warnings about real side effects. VAERS data helped to identify allergic reactions and blood clotting disorders caused by the COVID-19 vaccines. Both side effects are extremely rare, and doctors say the benefits of vaccination far outweigh the risks.

"While VAERS has limitations, keeping the system open to all reporters and users is essential for VAERS to serve its early detection function," the agency says.

Orenstein says he agrees with keeping VAERS as open as possible. "My feeling is that this is what we have to live with," he says, "because I think it's very important that we have an open and transparent system."

Even though some anti-vaccine advocates will distort the data, he says he thinks it's better to have it out there — available for any member of the public to see.

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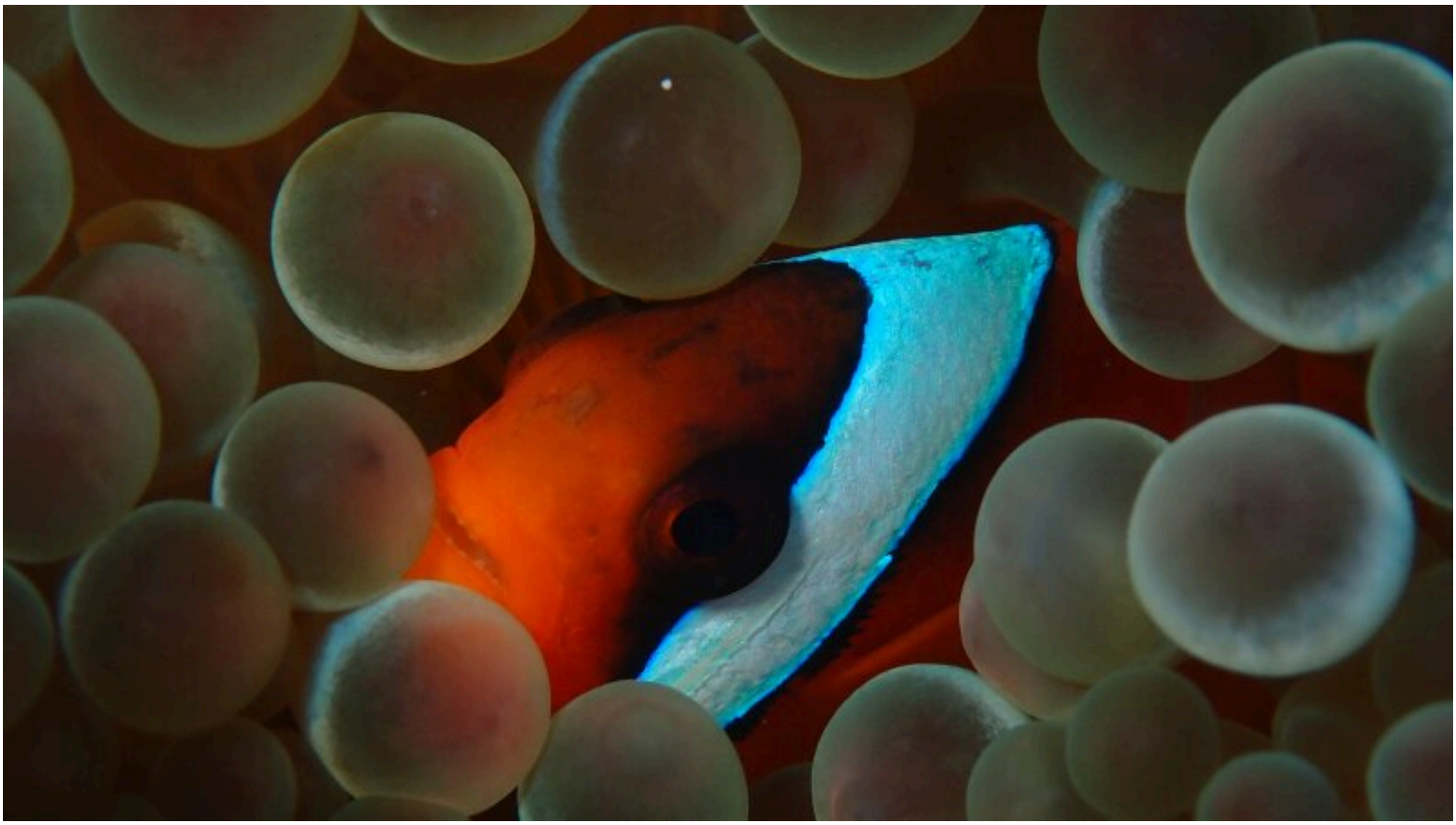

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